

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-14

Course: EEE F311 Communication Systems
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Date: 26-11-2025

1 Objectives

- Source coding and Mutual Information

2 Tutorial Problems

1. Q1 TB 13.1-1 Appended

13.1-1 A message source generates one of four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2, and 0.1. Each emitted message is independent of the other messages in the sequence.

(a) What is the source entropy?

(b) What is the rate of information generated by this source (in bits per second)?

Solution:

12.1-1 $P_1 = 0.4$, $P_2 = 0.3$, $P_3 = 0.2$, and $P_4 = 0.1$

$$H(m) = -(P_1 \log P_1 + P_2 \log P_2 + P_3 \log P_3 + P_4 \log P_4) = 1.846 \text{ bits (source entropy)}$$

There are 10^6 symbols/s. Hence, the rate of information generation is 1.846×10^6 bits/s

2. Q2 TB 13.2-1 Appended

13.2-1 A source emits seven messages with probabilities $1/2$, $1/4$, $1/8$, $1/16$, $1/32$, $1/64$, and $1/64$, respectively. Find the entropy of the source. Obtain the compact binary code and find the average length of the codeword. Determine the efficiency and the redundancy of the code.

Solution:

12.2-1

$$H(m) = \sum_{i=1}^7 P_i \log P_i = \frac{63}{32} \text{ bits}$$

$$L = \sum_i P_i L_i = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + \frac{1}{32}(5) + \frac{1}{64}(6) + \frac{1}{64}(6) = \frac{63}{32} \text{ binarydigits}$$

Efficiency $\eta = \frac{H(m)}{L} \times 100 = 100\%$ Redundancy $\gamma = (100 - \eta) = 0\%$. See Table S12.2-1 for the codewords.

Table S12.2-1

Message	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code
m_1	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	0
m_2	$1/4$	10	$1/4$	10	$1/4$	10	$1/4$	10	$1/4$	10	$\rightarrow 1/2$	1
m_3	$1/8$	110	$1/8$	110	$1/8$	110	$1/8$	110	$\rightarrow 1/4$	11		
m_4	$1/16$	1110	$1/16$	1110	$1/16$	1110	$\rightarrow 1/8$	111				
m_5	$1/32$	11110	$1/32$	11110	$\rightarrow 1/16$	1111						
m_6	$1/64$	111110	$\rightarrow 1/32$	11111								
m_7	$1/64$	111111										

3. Q3 TB 13.4-1 Appended

13.4-1 A binary channel matrix is given by

$$\begin{array}{cc} & \text{Outputs} \\ & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} \text{Inputs} \\ x_1 \\ x_2 \end{matrix} & \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{9}{10} \end{pmatrix} \end{array}$$

This means $P_{y|x}(y_1|x_1) = 2/3$, $P_{y|x}(y_2|x_1) = 1/3$, etc. You are also given that $P_x(x_1) = 1/3$ and $P_x(x_2) = 2/3$. Determine $H(x)$, $H(x|y)$, $H(y)$, $H(y|x)$, and $I(x; y)$.

Solution:

12.4-1

(a)

$$P(y_1) = P(y_1|x_1)P(x_1) + P(y_1|x_2)P(x_2) = \frac{2}{3} \frac{1}{3} + \frac{1}{10} \frac{2}{3} = \frac{13}{45}$$

$$P(y_2) = 1 - P(y_1) = \frac{32}{45}$$

(b)

$$H(x) = P(x_1) \log \frac{1}{P(x_1)} + P(x_2) \log \frac{1}{P(x_2)} = \frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 \frac{3}{2} = 0.918 \text{ bits}$$

To compute $H(x|y)$, we find

$$P(x_1|y_1) = \frac{P(y_1|x_1)P(x_1)}{P(y_1)} = \frac{10}{13}$$

$$P(x_1|y_2) = \frac{P(y_2|x_1)P(x_1)}{P(y_2)} = \frac{5}{32}$$

$$P(x_2|y_1) = \frac{P(y_1|x_2)P(x_2)}{P(y_1)} = \frac{3}{13}$$

$$P(x_2|y_2) = \frac{P(y_2|x_2)P(x_2)}{P(y_2)} = \frac{54}{64}$$

$$H(x|y_1) = P(x_1|y_1) \log \frac{1}{P(x_1|y_1)} + P(x_2|y_1) \log \frac{1}{P(x_2|y_1)} = \frac{10}{13} \log_2 \frac{13}{10} + \frac{3}{13} \log_2 \frac{13}{3} = 0.779$$

$$H(x|y_2) = P(x_1|y_2) \log \frac{1}{P(x_1|y_2)} + P(x_2|y_2) \log \frac{1}{P(x_2|y_2)} = \frac{5}{32} \log_2 \frac{32}{5} + \frac{54}{64} \log_2 \frac{64}{54} = 0.624$$

and

$$H(x|y) = P(y_1)H(x|y_1) + P(y_2)H(x|y_2) = \frac{13}{45}(0.779) + \frac{32}{45}(0.624) = 0.6687$$

Thus,

$$I(x|y) = H(x) - H(x|y) = 0.918 - 0.6687 = 0.2493 \text{ bits/binit}$$

$$H(y) = \sum_i P(y_i) \log \frac{1}{P(y_i)} = \frac{13}{45} \log \frac{45}{13} + \frac{32}{45} \log \frac{45}{32} = 0.8673 \text{ bits/symbol}$$

Also,

$$H(y|x) = H(y) - I(x|y) = 0.8673 - 0.2493 = 0.618 \text{ bits/symbol}$$